BA Capstone Project: Checkpoint 1

Data Fundamentals & SQL Querying

Group Name: MANTA RAY
Course & Section: BED 106 - Business Analytics
Date: [SUBMISSION DATE]

Team Members & Roles:

- Sugar-Rey R. Garcia — Project Lead / Analyst
- Rudelyn Joy Abanador— Data Engineer
- Joel Madula — Statistician / Modeler

Task 1.1 — Business Problem Statement

Business Domain & Problem Investigation

This project belongs to the Retail & Sales Analytics domain. The business problem being investigated is Sales Performance & Revenue Forecasting — specifically, identifying which products and regions generate the most revenue, analyzing sales trends over time, and forecasting future revenue to support business decision-making. Using historical transactional data from a retail superstore, this project aims to uncover patterns in sales, profit, and discount behavior across different product categories and geographic regions.

Business Impact & Significance

Understanding sales performance is critical for any retail business. Without clear visibility into which products, regions, and time periods drive the most revenue, businesses risk over-investing in low-performing areas and missing growth opportunities. Accurate revenue forecasting enables management to plan inventory, allocate budgets, and set realistic sales targets. This project directly addresses the need for data-driven decision-making in retail operations, which can lead to improved profitability, reduced waste from overstocking, and more targeted marketing efforts toward high-value customer segments.

Key Business Questions

1. Which product categories and regions generate the highest total sales and profit?
2. Does offering higher discounts lead to lower profit margins?
3. What are the monthly and seasonal sales trends over time?
4. Which customer segments contribute the most to total revenue?
5. What will next quarter's sales look like based on historical trends?

Data Source Overview

The dataset planned for this project is the Sample Superstore Sales Dataset, a publicly available retail transaction dataset sourced from Kaggle (www.kaggle.com). It contains structured, tabular data consisting of 9,994 rows of order-level records with 21 columns covering customer information, geographic data, product categories, and financial metrics such as sales, profit, discount, and quantity. The data represents multiple years of retail transactions, making it suitable for time-series trend analysis and regression-based forecasting.

Task 1.2 — Dataset Documentation

Data Source Report

Attribute	Details
Dataset Name	Sample Superstore Sales Dataset
Source URL/File	https://www.kaggle.com/datasets/vivek468/superstore-dataset-final
Date Accessed	19/07/2026
License Type	CC0: Public Domain (No restrictions on use)

Data Dictionary

Column Name	Data Type	Description
Order ID	VARCHAR(20)	Unique identifier for each order transaction
Order Date	DATE	The date when the order was placed
Ship Date	DATE	The date when the order was shipped
Ship Mode	VARCHAR(30)	Shipping method selected (e.g., First Class, Standard Class)
Customer ID	VARCHAR(20)	Unique identifier for each customer
Customer Name	VARCHAR(100)	Full name of the customer
Segment	VARCHAR(30)	Customer market segment (Consumer, Corporate, Home Office)
Country	VARCHAR(50)	Country where the order was delivered
City	VARCHAR(50)	City where the order was delivered
State	VARCHAR(50)	State where the order was delivered
Region	VARCHAR(20)	Geographic region (East, West, Central, South)
Product ID	VARCHAR(20)	Unique identifier for each product

Column Name	Data Type	Description
Category	VARCHAR(30)	Main product category (Furniture, Office Supplies, Technology)
Sub-Category	VARCHAR(30)	Product sub-category (e.g., Chairs, Phones, Binders)
Product Name	VARCHAR(200)	Full descriptive name of the product
Sales	DECIMAL(10,2)	Total sales revenue for the order line item
Quantity	INT	Number of units ordered
Discount	DECIMAL(5,2)	Discount rate applied to the order (0.0 to 1.0)
Profit	DECIMAL(10,2)	Net profit earned from the order line item

Preliminary Data Quality Assessment

- **Missing Values:** *The Sample Superstore dataset has no missing values across all 9,994 rows and 21 columns. All fields are fully populated with no null or empty entries detected.*

- **Duplicates:** *No exact duplicate rows were found. However, the same Order ID may appear multiple times when a single order contains multiple products, which is expected and normal behavior in order-line transactional datasets.*

- **Inconsistencies:** *Dates are stored in MM/DD/YYYY format requiring conversion to YYYY-MM-DD for MySQL. Postal Code contains numeric values stored as text which must be handled as VARCHAR. A number of records show negative Profit* *values indicating loss-making orders — these are valid data points and are retained for analysis.*

- **Cleaning Steps Applied:** *Date columns (Order Date, Ship Date) were reformatted to YYYY-MM-DD for MySQL compatibility. Postal Code was cast as VARCHAR. Column names were standardized by removing spaces and special characters. The flat dataset was split into three relational tables: customers, products, and orders, with primary keys and foreign keys assigned accordingly.*

Raw Dataset Preview

The first 10–20 rows of the raw Superstore CSV file

Screenshot 1 – [*https://www.kaggle.com/datasets/vivek468/superstore-dataset-final*](https://www.kaggle.com/datasets/vivek468/superstore-dataset-final)

Sample - Superstore.csv (2.29 MB)

Detail Compact Column

21 of 21 columns

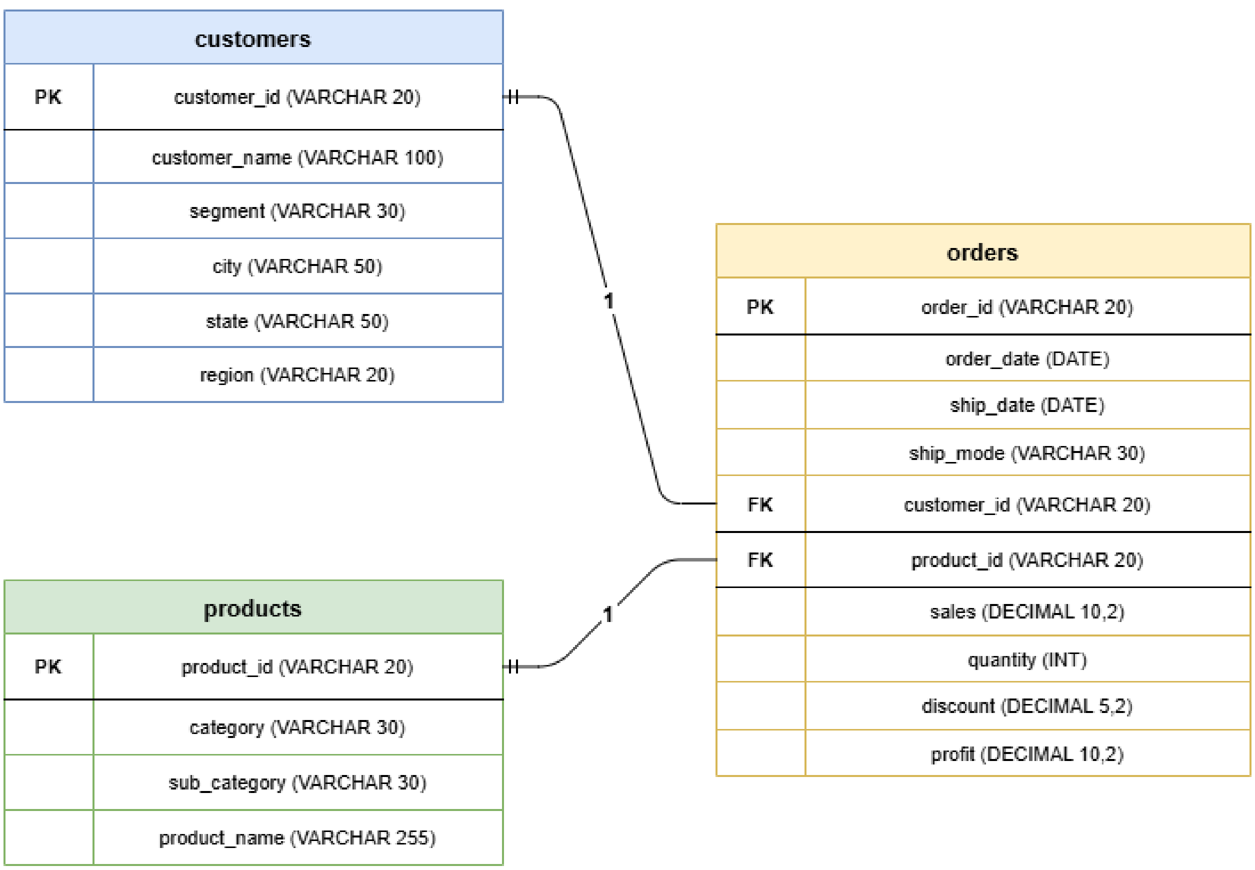
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ...	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Categ...	Product N...	Sales	Quantity	Discount
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.96	2	0
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	3	0
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by Universal	14.62	2	0
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	0.45
5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	22.368	2	0.2
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	7	0
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322	7.28	4	0
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone	907.152	6	0.2
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by Samsill	18.504	3	0.2
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge	114.9	5	0
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tables	1706.184	9	0.2
12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002033	Technology	Phones	Konftel 250 Conference phone - Charcoal black	911.424	4	0.2
13	CA-2017-114412	4/15/2017	4/20/2017	Standard Class	AA-10480	Andrew Allen	Consumer	United States	Concord	North Carolina	28027	South	OFF-PA-10002365	Office Supplies	Paper	Xerox 1967	15.552	3	0.2
14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-10003656	Office Supplies	Binders	Fellowes PB200 Plastic Comb Binding Machine	407.976	3	0.2
15	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-AP-10002311	Office Supplies	Appliances	Holmes Replacement	68.81	5	0.8

Screenshot 2 – Downloaded CSV file (Sample – Superstore.csv)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	Row ID	Order ID	Order Date	Ship Date	Ship Mod	Customer	Customer	Segment	Country	City	State	Postal Co	Region	Product ID	Category	Sub-Categ	Product N	Sales	Quantity	Discount	Profit
2	1	CA-2016-1	11/08/2016	11/11/2016	Second Cl	CG-12520	Claire Gu	Consume	United St	Hendersc	Kentucky	42420	South	FUR-BO-1	Furniture	Bookcase	Bush Som	261.96	2	0	41.9136
3	2	CA-2016-1	11/08/2016	11/11/2016	Second Cl	CG-12520	Claire Gu	Consume	United St	Hendersc	Kentucky	42420	South	FUR-CH-1	Furniture	Chairs	Hon Delu	731.94	3	0	219.582
4	3	CA-2016-1	06/12/2016	6/16/2016	Second Cl	DV-13045	Darrin Va	Corporate	United St	Los Angel	California	90036	West	OFF-LA-10	Office Sup	Labels	Self-Adhe	14.62	2	0	6.8714
5	4	US-2015-1	10/11/2015	10/18/2015	Standard	SO-20335	Sean O'D	Consume	United St	Fort Laud	Florida	33311	South	FUR-TA-1	Furniture	Tables	Bretford C	957.5775	5	0.45	-383.031
6	5	US-2015-1	10/11/2015	10/18/2015	Standard	SO-20335	Sean O'D	Consume	United St	Fort Laud	Florida	33311	South	OFF-ST-10	Office Sup	Storage	Eldon Fol	22.368	2	0.2	2.5164
7	6	CA-2014-1	06/09/2014	6/14/2014	Standard	BH-11710	Brosina H	Consume	United St	Los Angel	California	90032	West	FUR-FU-1	Furniture	Furnishin	Eldon Exp	48.86	7	0	14.1694
8	7	CA-2014-1	06/09/2014	6/14/2014	Standard	BH-11710	Brosina H	Consume	United St	Los Angel	California	90032	West	OFF-AR-1	Office Sup	Art	Newell 32	7.28	4	0	1.9656
9	8	CA-2014-1	06/09/2014	6/14/2014	Standard	BH-11710	Brosina H	Consume	United St	Los Angel	California	90032	West	TEC-PH-10	Technolo	Phones	Mitel 532	907.152	6	0.2	90.7152
10	9	CA-2014-1	06/09/2014	6/14/2014	Standard	BH-11710	Brosina H	Consume	United St	Los Angel	California	90032	West	OFF-BI-10	Office Sup	Binders	DXL Angle	18.504	3	0.2	5.7825
11	10	CA-2014-1	06/09/2014	6/14/2014	Standard	BH-11710	Brosina H	Consume	United St	Los Angel	California	90032	West	OFF-AP-1	Office Sup	Appliance	Belkin F5	114.9	5	0	34.47
12	11	CA-2014-1	06/09/2014	6/14/2014	Standard	BH-11710	Brosina H	Consume	United St	Los Angel	California	90032	West	FUR-TA-1	Furniture	Tables	Chromcra	1706.184	9	0.2	85.3092
13	12	CA-2014-1	06/09/2014	6/14/2014	Standard	BH-11710	Brosina H	Consume	United St	Los Angel	California	90032	West	TEC-PH-10	Technolo	Phones	Konftel 25	911.424	4	0.2	68.3568
14	13	CA-2017-1	4/15/2017	4/20/2017	Standard	AA-10480	Andrew A	Consume	United St	Concord	North Car	28027	South	OFF-PA-1	Office Sup	Paper	Xerox 196	15.552	3	0.2	5.4432
15	14	CA-2016-1	12/05/2016	12/10/2016	Standard	IM-15070	Irene Ma	Consume	United St	Seattle	Washingt	98103	West	OFF-BI-10	Office Sup	Binders	Fellowes	407.976	3	0.2	132.5922
16	15	US-2015-1	11/22/2015	11/26/2015	Standard	HP-14815	Harold Pa	Home Off	United St	Fort Wort	Texas	76106	Central	OFF-AP-1	Office Sup	Appliance	Holmes R	68.81	5	0.8	-123.858
17	16	US-2015-1	11/22/2015	11/26/2015	Standard	HP-14815	Harold Pa	Home Off	United St	Fort Wort	Texas	76106	Central	OFF-BI-10	Office Sup	Binders	Storex Du	2.544	3	0.8	-3.816
18	17	CA-2014-1	11/11/2014	11/18/2014	Standard	PK-19075	Pete Kriz	Consume	United St	Madison	Wisconsi	53711	Central	OFF-ST-10	Office Sup	Storage	Stur-D-Sto	665.88	6	0	13.3176
19	18	CA-2014-1	5/13/2014	5/15/2014	Second Cl	AG-10270	Alejandro	Consume	United St	West Jord	Utah	84084	West	OFF-ST-10	Office Sup	Storage	Fellowes	55.5	2	0	9.99
20	19	CA-2014-1	8/27/2014	09/01/2014	Second Cl	ZD-21925	Zuschuss	Consume	United St	San Franc	California	94109	West	OFF-AR-1	Office Sup	Art	Newell 34	8.56	2	0	2.4824

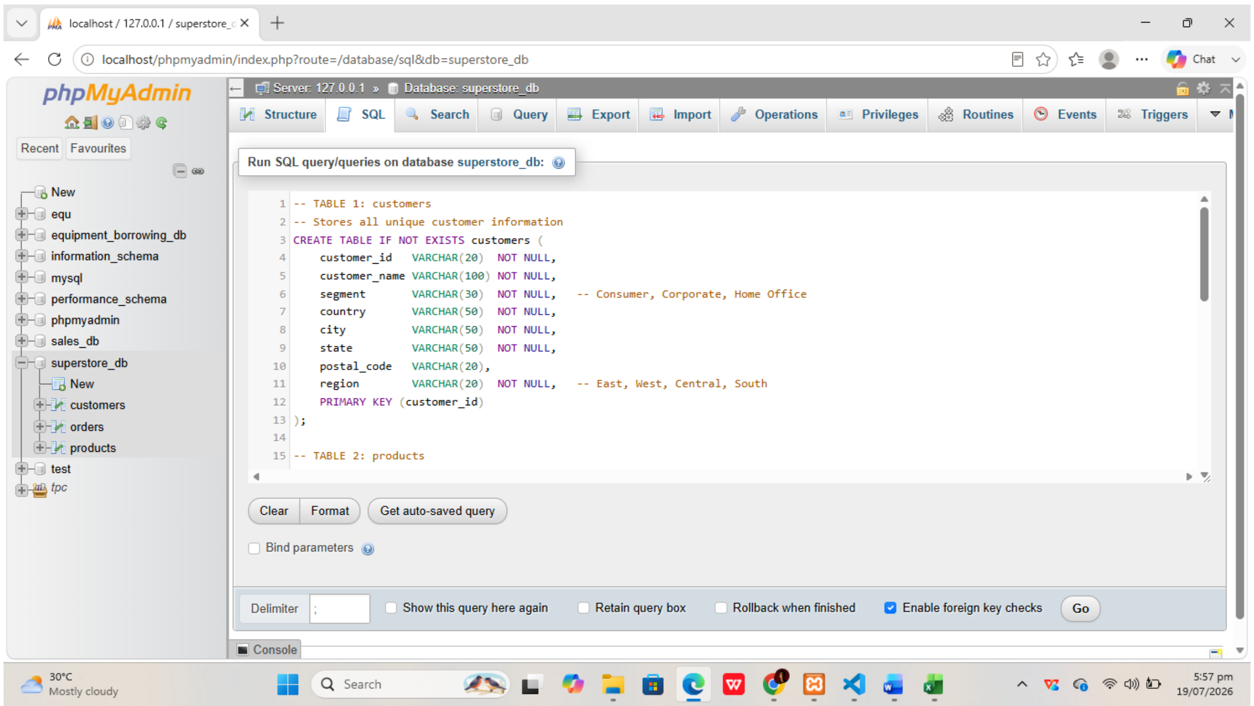
Task 1.3 — Relational Database Design

Entity-Relationship Diagram (ERD)

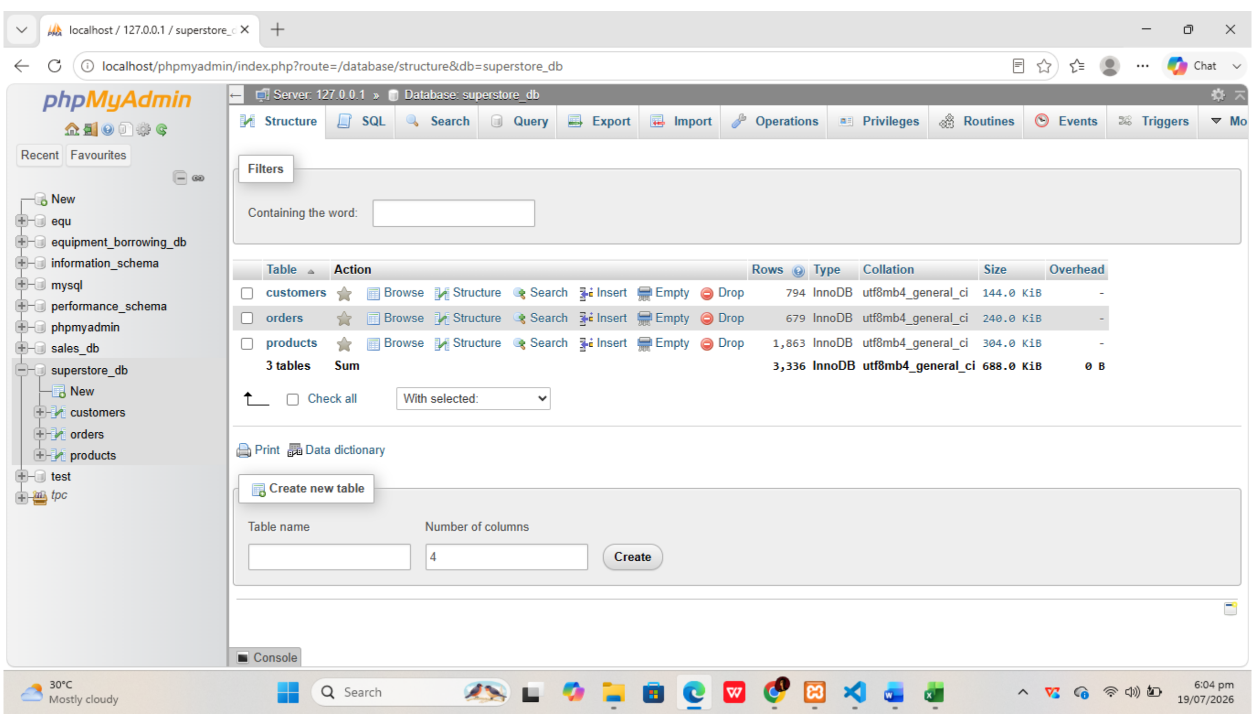


Database Schema & Population

Screenshot 1 — CREATE TABLE statements

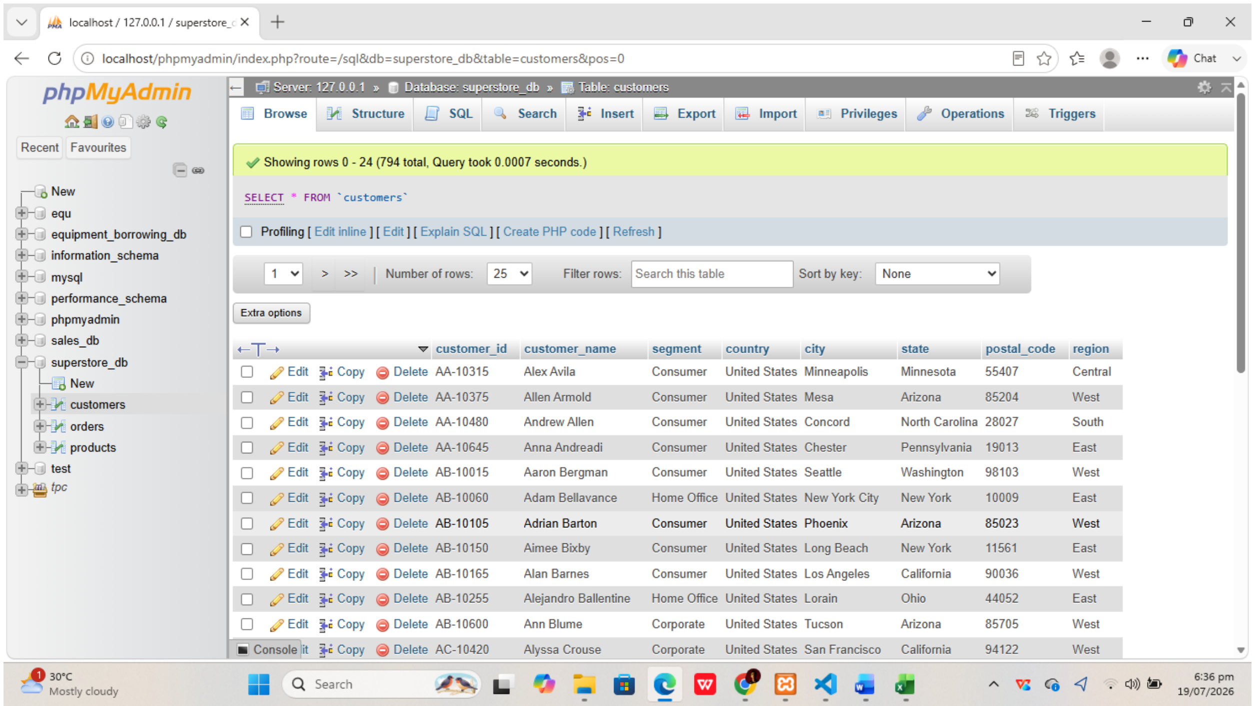


Screenshot 2 — Three tables in the schema



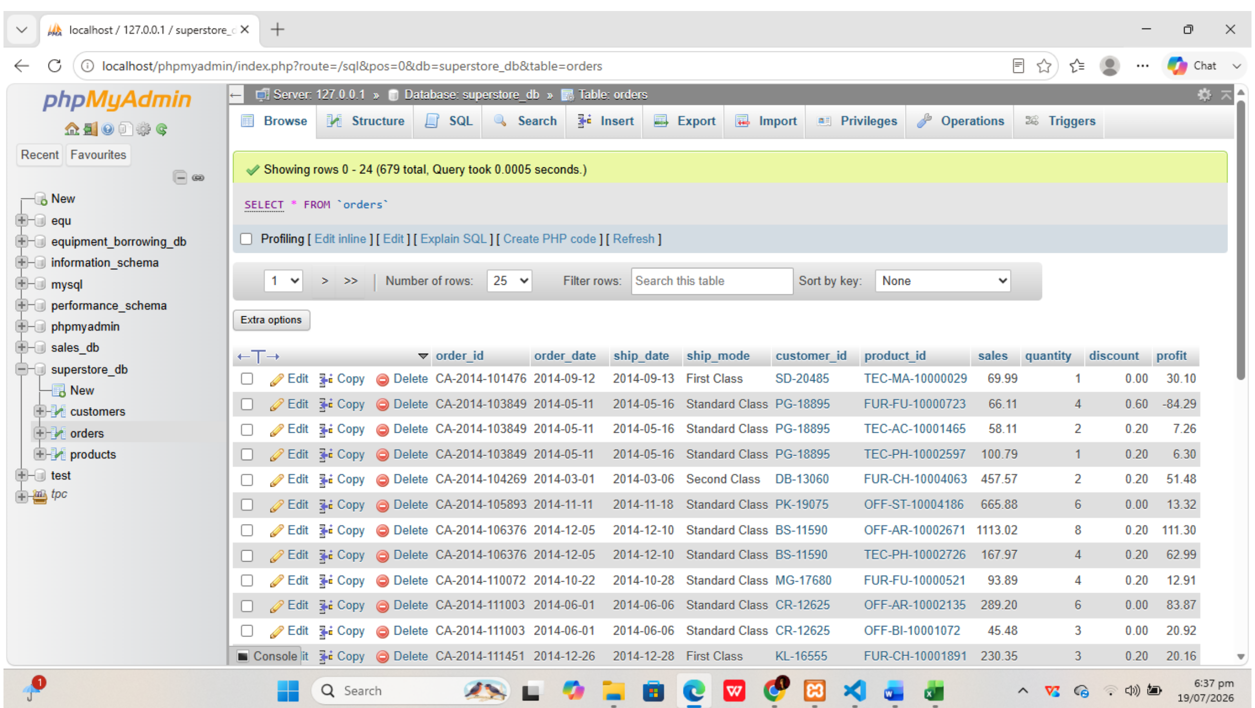
Screenshot 3 — Table previews

- ***Customers Table***



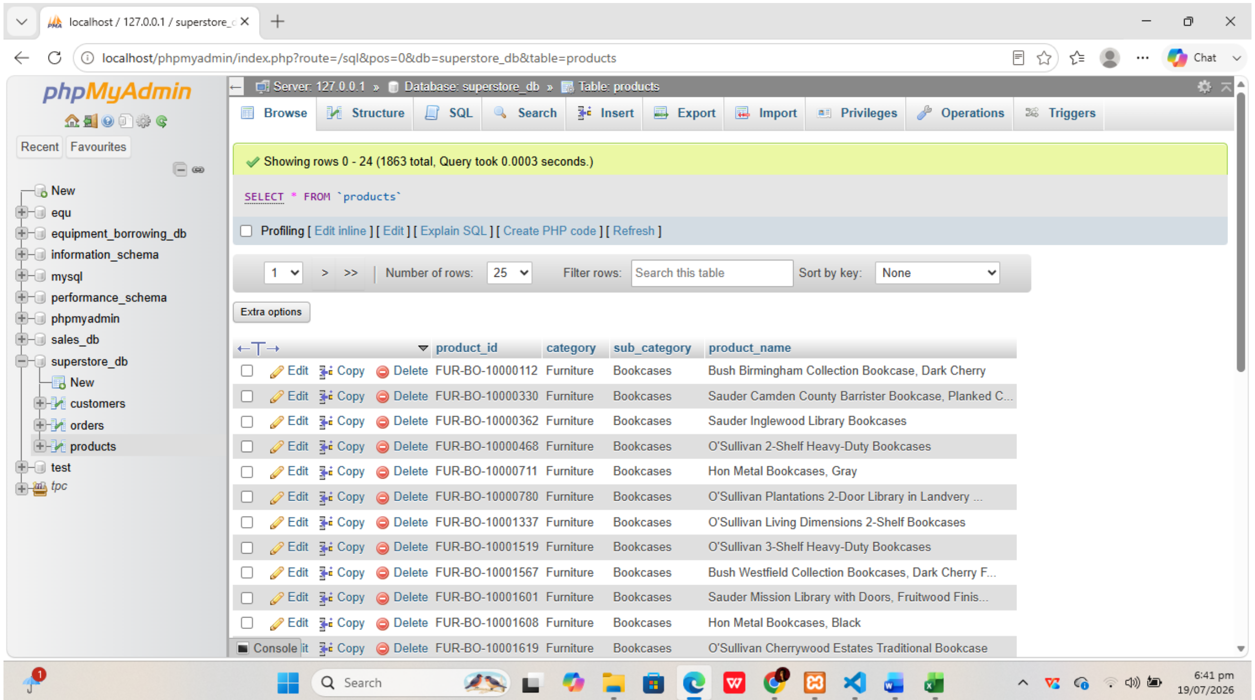
customer_id	customer_name	segment	country	city	state	postal_code	region
AA-10315	Alex Avila	Consumer	United States	Minneapolis	Minnesota	55407	Central
AA-10375	Allen Armold	Consumer	United States	Mesa	Arizona	85204	West
AA-10480	Andrew Allen	Consumer	United States	Concord	North Carolina	28027	South
AA-10645	Anna Andreadi	Consumer	United States	Chester	Pennsylvania	19013	East
AB-10015	Aaron Bergman	Consumer	United States	Seattle	Washington	98103	West
AB-10060	Adam Bellavance	Home Office	United States	New York City	New York	10009	East
AB-10105	Adrian Barton	Consumer	United States	Phoenix	Arizona	85023	West
AB-10150	Aimee Bixby	Consumer	United States	Long Beach	New York	11561	East
AB-10165	Alan Barnes	Consumer	United States	Los Angeles	California	90036	West
AB-10255	Alejandro Ballentine	Home Office	United States	Lorain	Ohio	44052	East
AB-10600	Ann Blume	Corporate	United States	Tucson	Arizona	85705	West
AC-10420	Alyssa Crouse	Corporate	United States	San Francisco	California	94122	West

- ***Orders Table***



order_id	order_date	ship_date	ship_mode	customer_id	product_id	sales	quantity	discount	profit
CA-2014-101476	2014-09-12	2014-09-13	First Class	SD-20485	TEC-MA-10000029	69.99	1	0.00	30.10
CA-2014-103849	2014-05-11	2014-05-16	Standard Class	PG-18895	FUR-FU-10000723	66.11	4	0.60	-84.29
CA-2014-103849	2014-05-11	2014-05-16	Standard Class	PG-18895	TEC-AC-10001465	58.11	2	0.20	7.26
CA-2014-103849	2014-05-11	2014-05-16	Standard Class	PG-18895	TEC-PH-10002597	100.79	1	0.20	6.30
CA-2014-104269	2014-03-01	2014-03-06	Second Class	DB-13060	FUR-CH-10004063	457.57	2	0.20	51.48
CA-2014-105893	2014-11-11	2014-11-18	Standard Class	PK-19075	OFF-ST-10004186	665.88	6	0.00	13.32
CA-2014-106376	2014-12-05	2014-12-10	Standard Class	BS-11590	OFF-AR-10002671	1113.02	8	0.20	111.30
CA-2014-106376	2014-12-05	2014-12-10	Standard Class	BS-11590	TEC-PH-10002726	167.97	4	0.20	62.99
CA-2014-110072	2014-10-22	2014-10-28	Standard Class	MG-17680	FUR-FU-10000521	93.89	4	0.20	12.91
CA-2014-111003	2014-06-01	2014-06-06	Standard Class	CR-12625	OFF-AR-10002135	289.20	6	0.00	83.87
CA-2014-111003	2014-06-01	2014-06-06	Standard Class	CR-12625	OFF-BI-10001072	45.48	3	0.00	20.92
CA-2014-111451	2014-12-26	2014-12-28	First Class	KL-16555	FUR-CH-10001891	230.35	3	0.20	20.16

- ***Products Table***



Task 1.4 — SQL Query Report

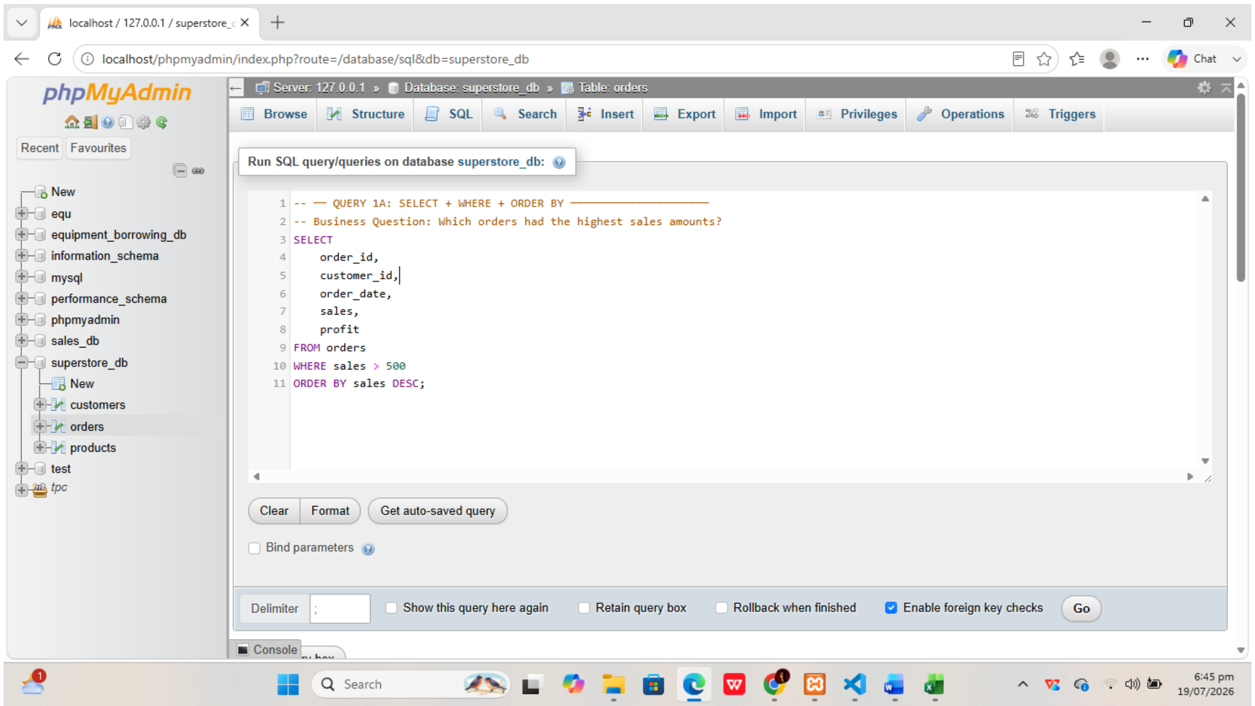
1. Basic Data Retrieval (2 Queries)

Must include SELECT, WHERE, and ORDER BY clauses.

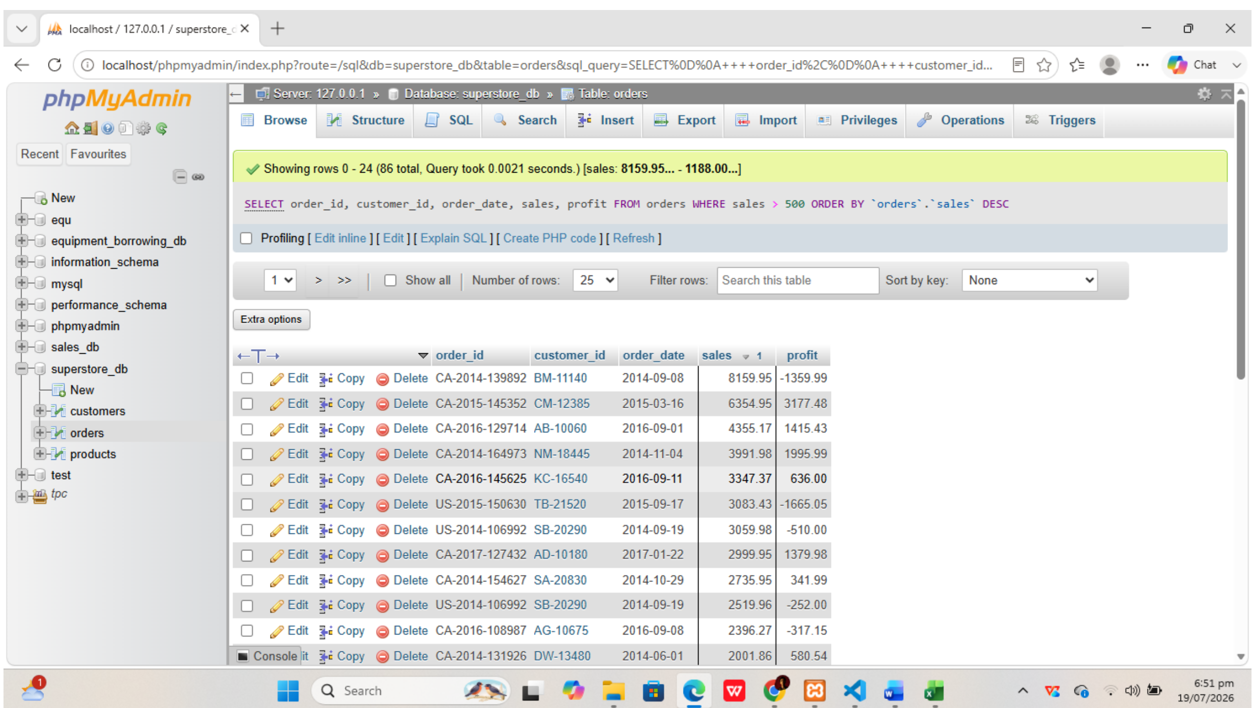
QUERY 1B: SELECT + WHERE + ORDER BY

- *-- Business Question: What orders were placed in 2017?*

- ***Screenshot of SQL Code***



- ***Result Screenshot***



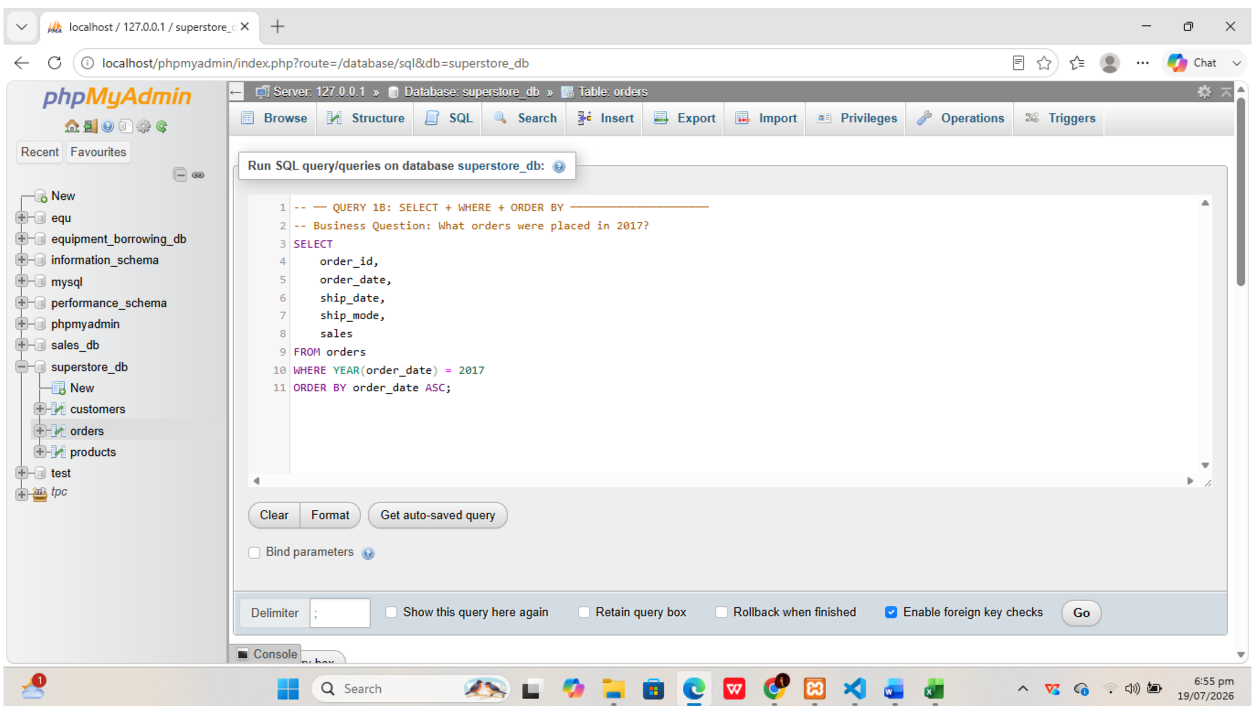
- ***Business Interpretation***

  Query 1A: Retrieves all orders where the sales amount exceeds 500, sorted from highest to lowest. This identifies high-value transactions that contribute most significantly to total revenue, helping management understand patterns among large purchases such as which products or regions tend to produce high-value orders.

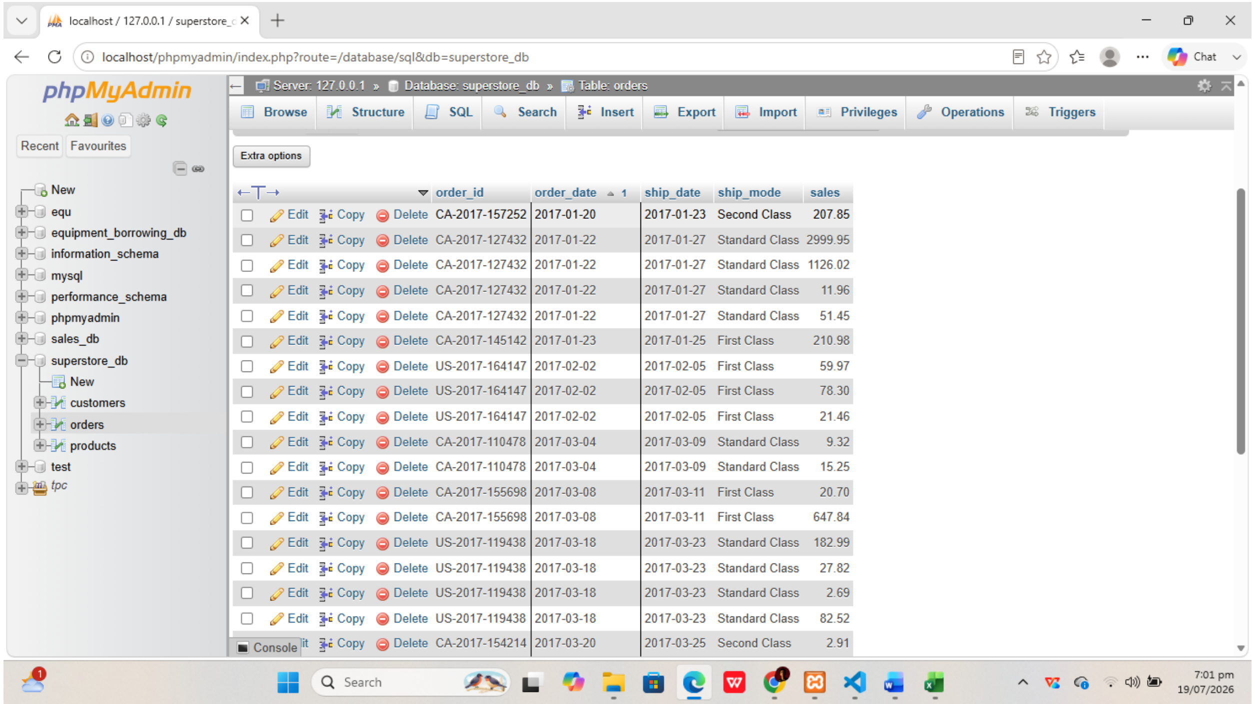
QUERY 1B: SELECT + WHERE + ORDER BY

-- Business Question: What orders were placed in 2017?

- ***Screenshot of SQL Code***



- ***Result Screenshot***



- **Business Interpretation:**

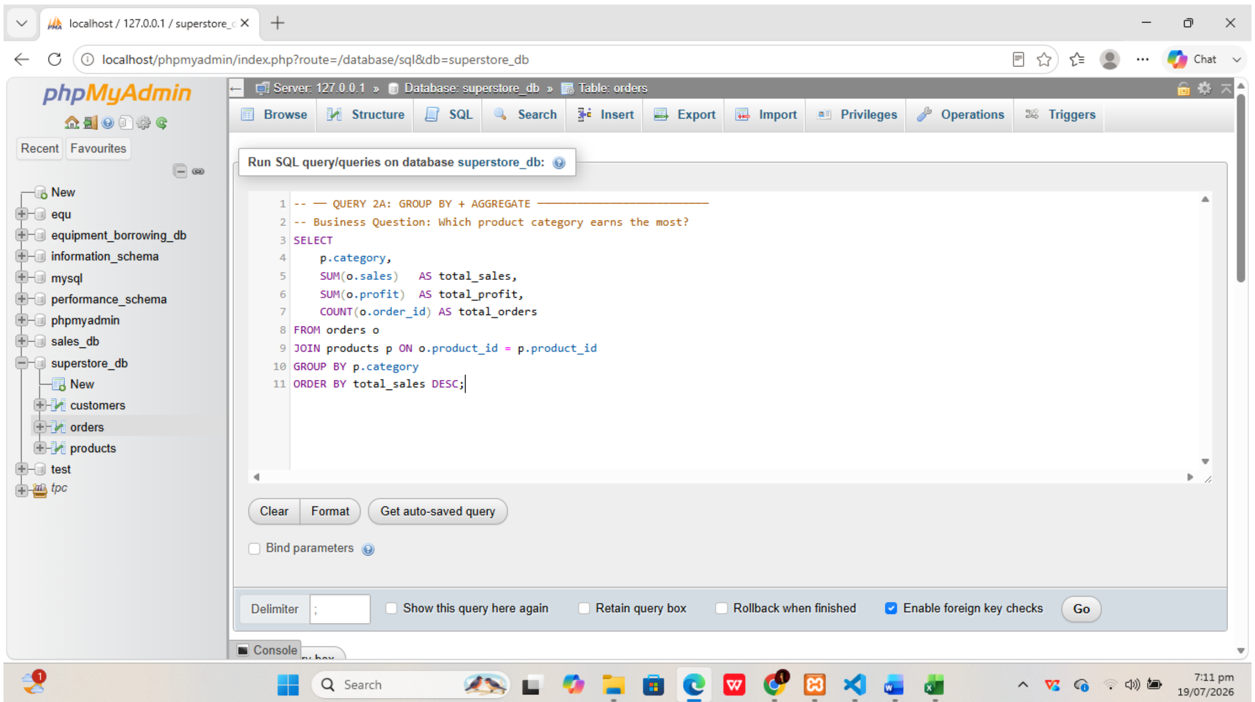
  Query 1B: Filters all orders placed in 2017 in chronological order, allowing the business to review the full transaction history for a specific fiscal year and compare annual sales volumes to track business growth over time.

2. Aggregate Analysis (2 Queries)

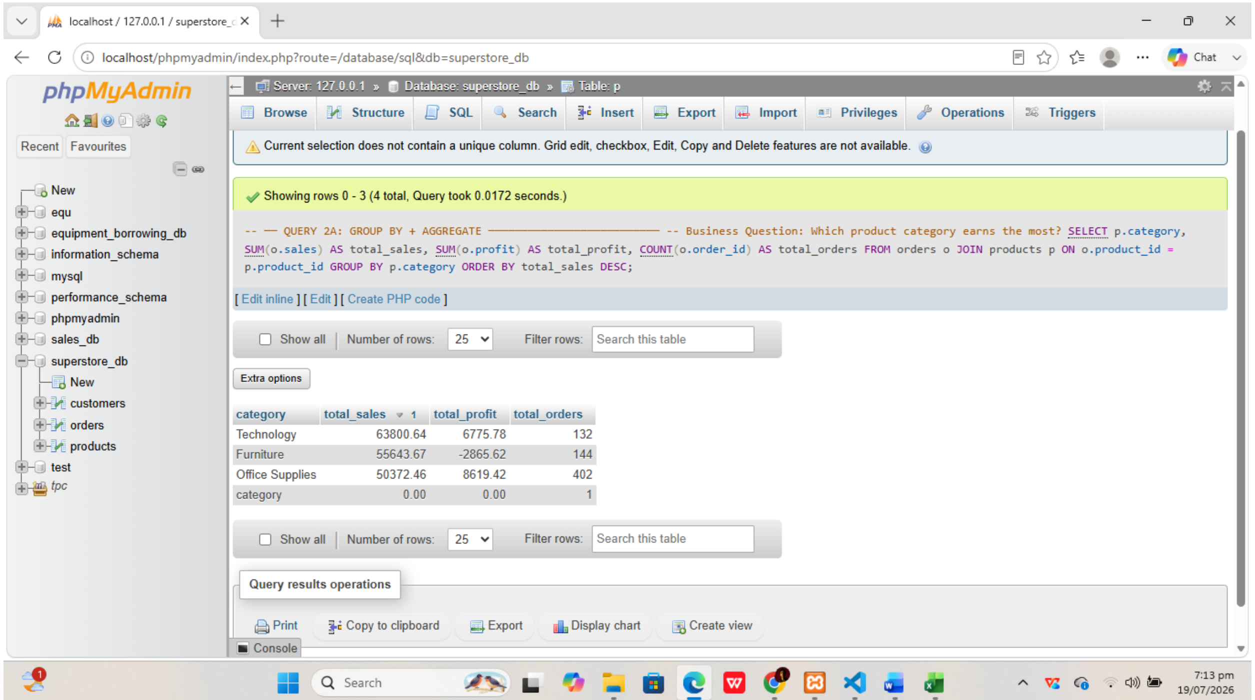
Must include GROUP BY and Aggregate Functions (COUNT, SUM, AVG, etc.).

QUERY 2A: GROUP BY + AGGREGATE
-- Business Question: Which product category earns the most?

- ***Screenshot of SQL Code***



- ***Result Screenshot***



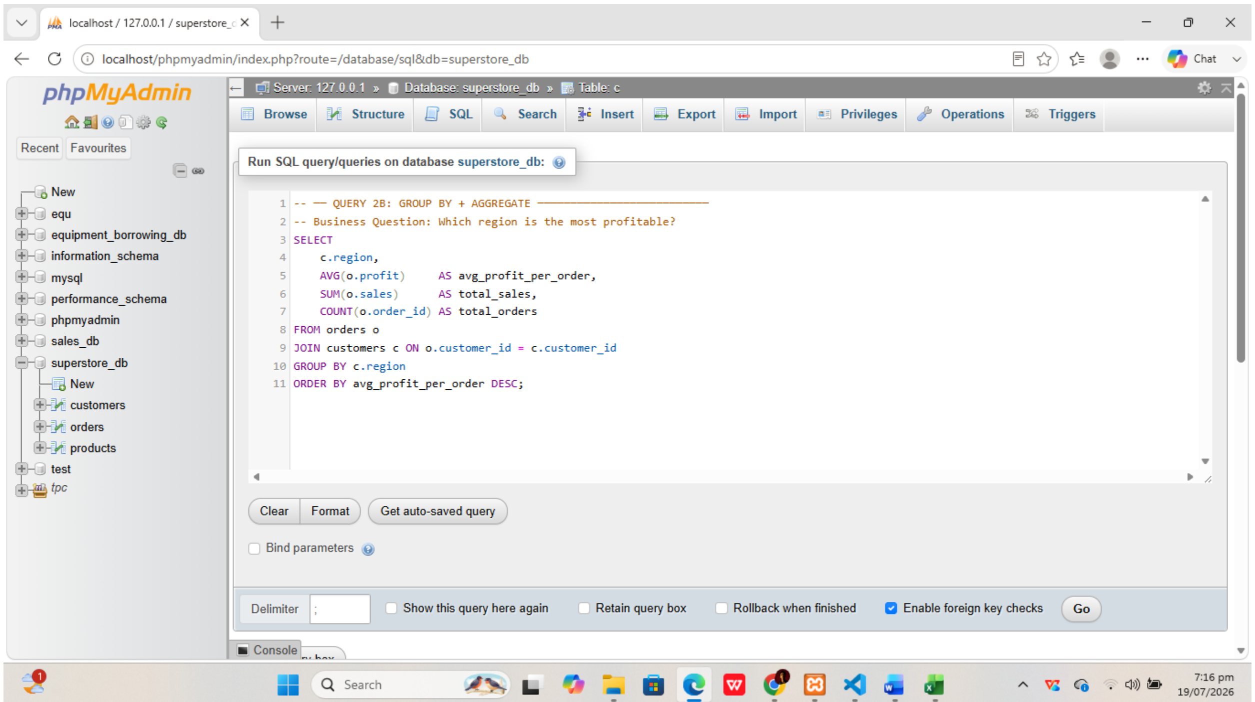
- **Business Interpretation:**

Query 2A: Calculates total sales revenue and total profit per product category. Results reveal which of the three main categories — Furniture, Office Supplies, or Technology — generates the most revenue and profit, informing product mix and resource allocation decisions.

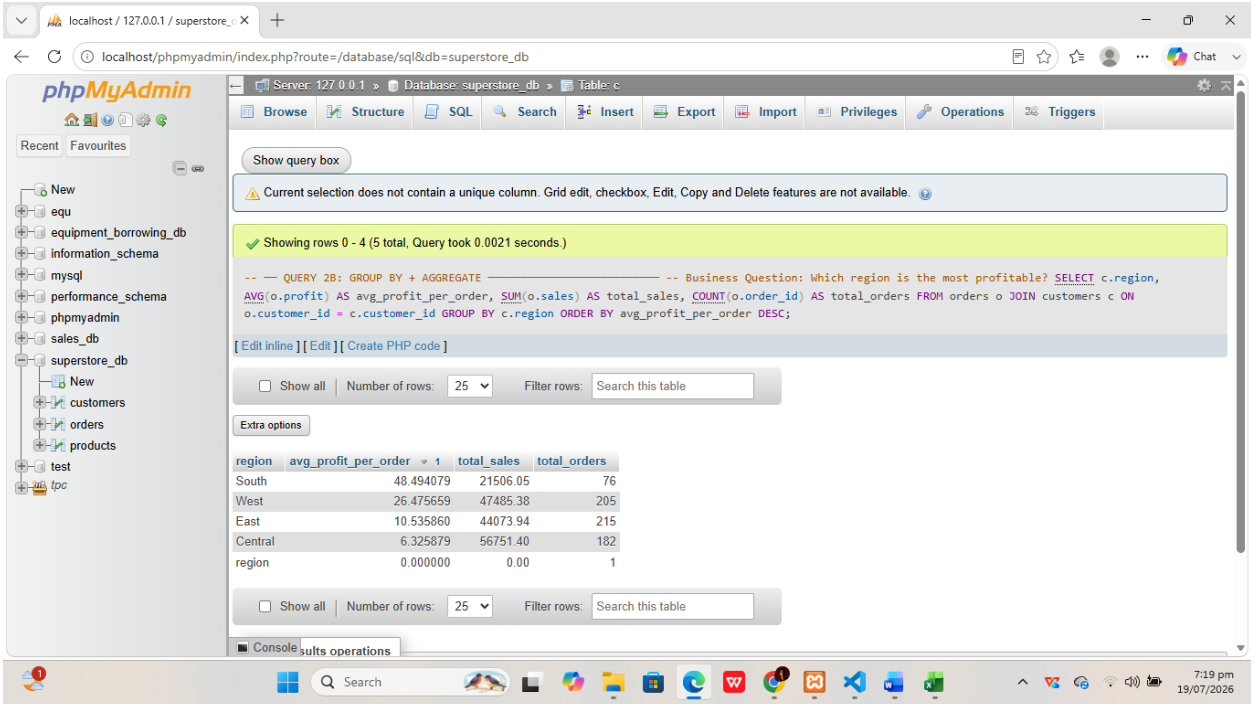
QUERY 2B: GROUP BY + AGGREGATE

-- Business Question: Which region is the most profitable?

- ***Screenshot of SQL Code***



- ***Result Screenshot***



- **Business Interpretation:**

Query 2B: Computes average profit per order and total order count by geographic region. A region with many orders but low average profit may indicate heavy discounting or inefficiencies, while a high-profit region signals a target for increased investment.

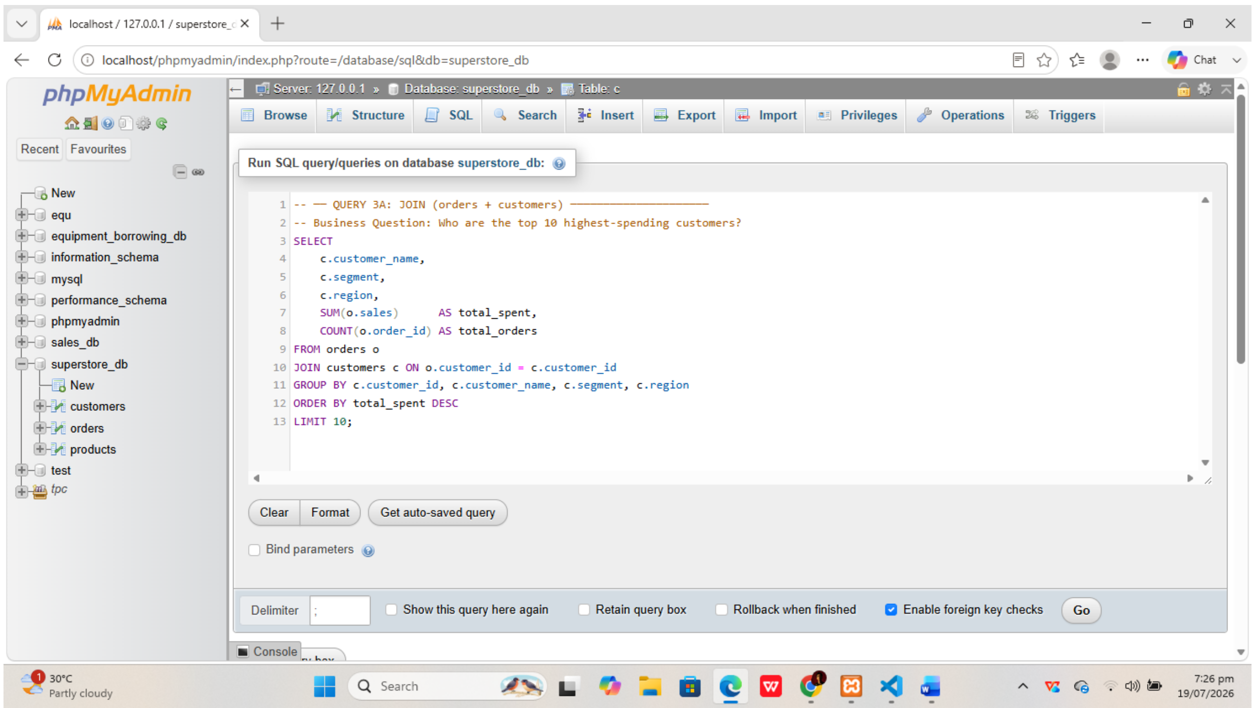
3. Multi-Table Joins (2 Queries)

Must connect two or more tables.

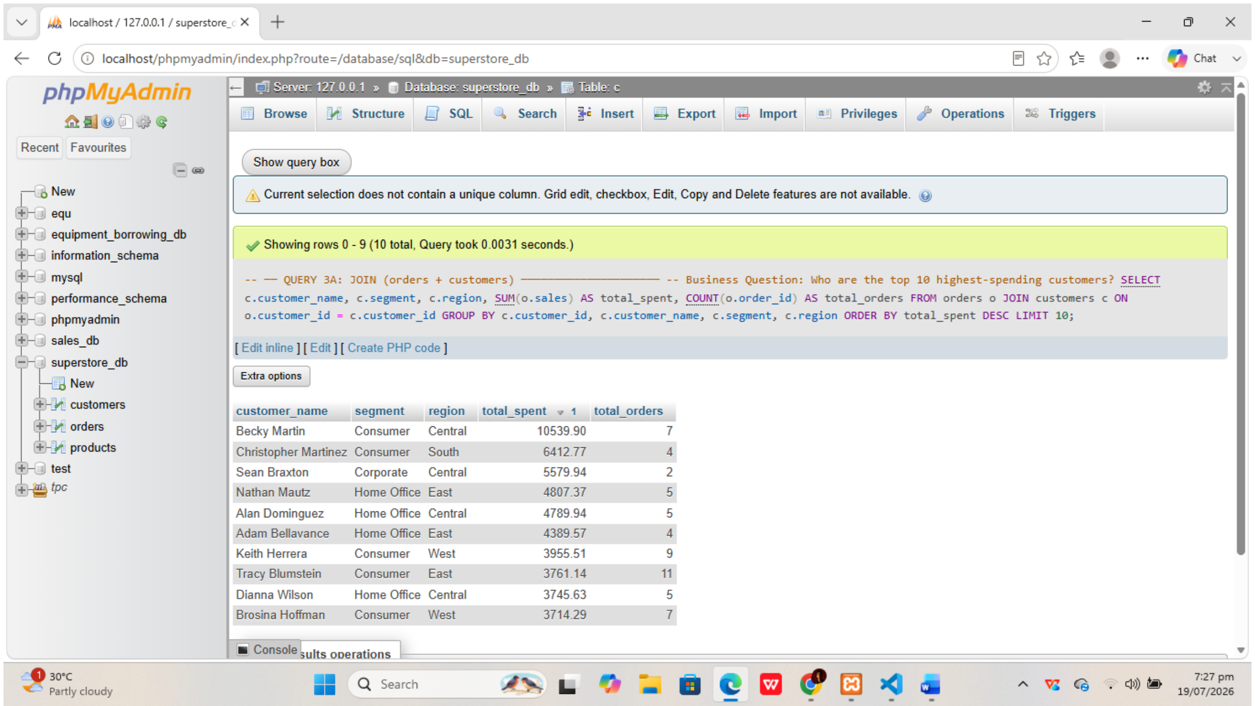
QUERY 3A: JOIN (orders + customers)

--Business Question: Who are the top 10 highest-spending customers?

- ***Screenshot of SQL Code***



- ***Result Screenshot***



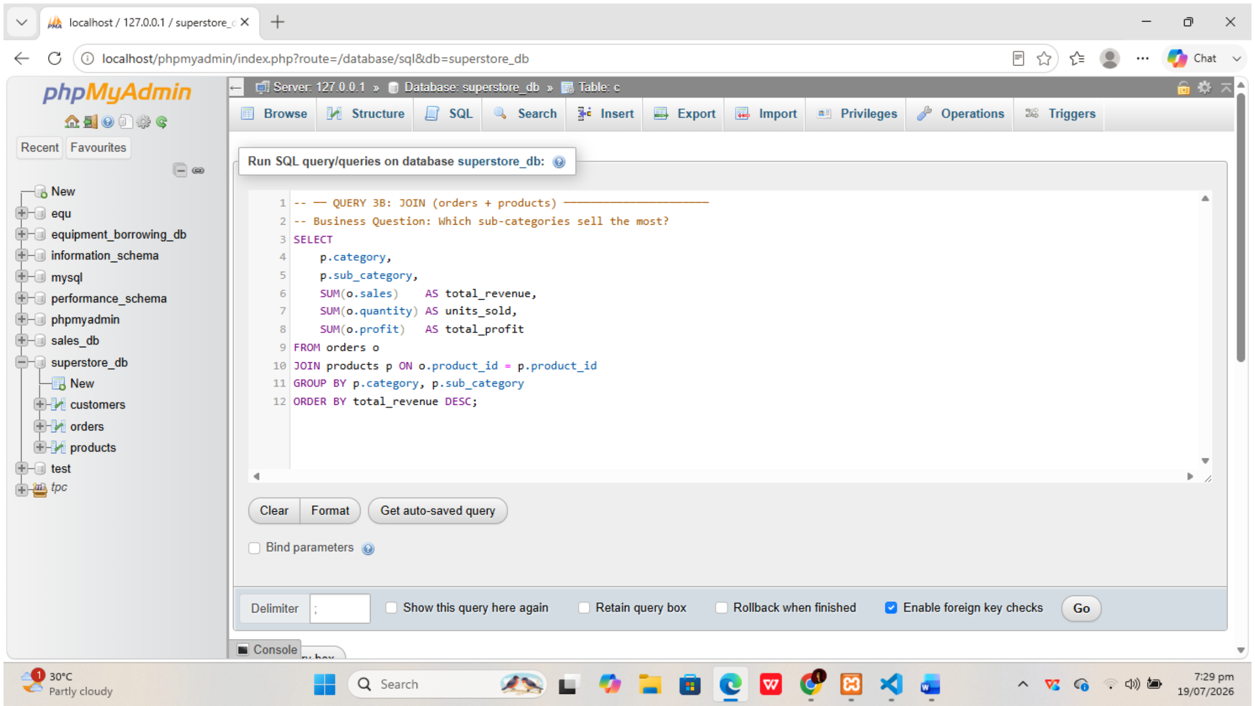
- ***Business Interpretation***

Query 3A: Joins orders and customers tables to identify the top 10 customers by total spending. Knowing which customers generate the most revenue enables the business to prioritize relationship management with high-value clients and design targeted loyalty programs.

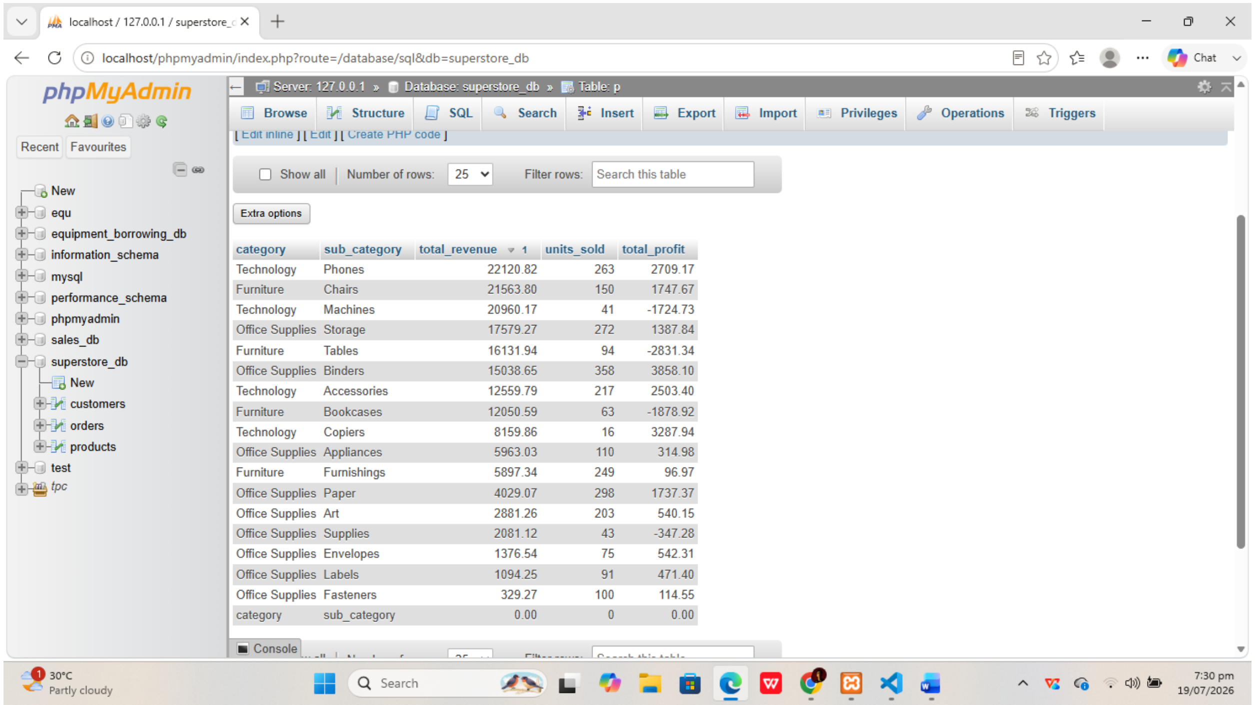
QUERY 3B: JOIN (orders + products)

-- Business Question: Which sub-categories sell the most?

- ***Screenshot of SQL Code***



- ***Result Screenshot***



category	sub_category	total_revenue	units_sold	total_profit
Technology	Phones	22120.82	263	2709.17
Furniture	Chairs	21563.80	150	1747.67
Technology	Machines	20960.17	41	-1724.73
Office Supplies	Storage	17579.27	272	1387.84
Furniture	Tables	16131.94	94	-2831.34
Office Supplies	Binders	15038.65	358	3858.10
Technology	Accessories	12559.79	217	2503.40
Furniture	Bookcases	12050.59	63	-1878.92
Technology	Copiers	8159.86	16	3287.94
Office Supplies	Appliances	5963.03	110	314.98
Furniture	Furnishings	5897.34	249	96.97
Office Supplies	Paper	4029.07	298	1737.37
Office Supplies	Art	2881.26	203	540.15
Office Supplies	Supplies	2081.12	43	-347.28
Office Supplies	Envelopes	1376.54	75	542.31
Office Supplies	Labels	1094.25	91	471.40
Office Supplies	Fasteners	329.27	100	114.55
category	sub_category	0.00	0	0.00

- ***Business Interpretation***

Query 2B: Joins orders and products tables to rank all sub-categories by total revenue and units sold, providing granular insight into which specific sub-categories like Phones, Chairs, or Binders are the strongest performers.

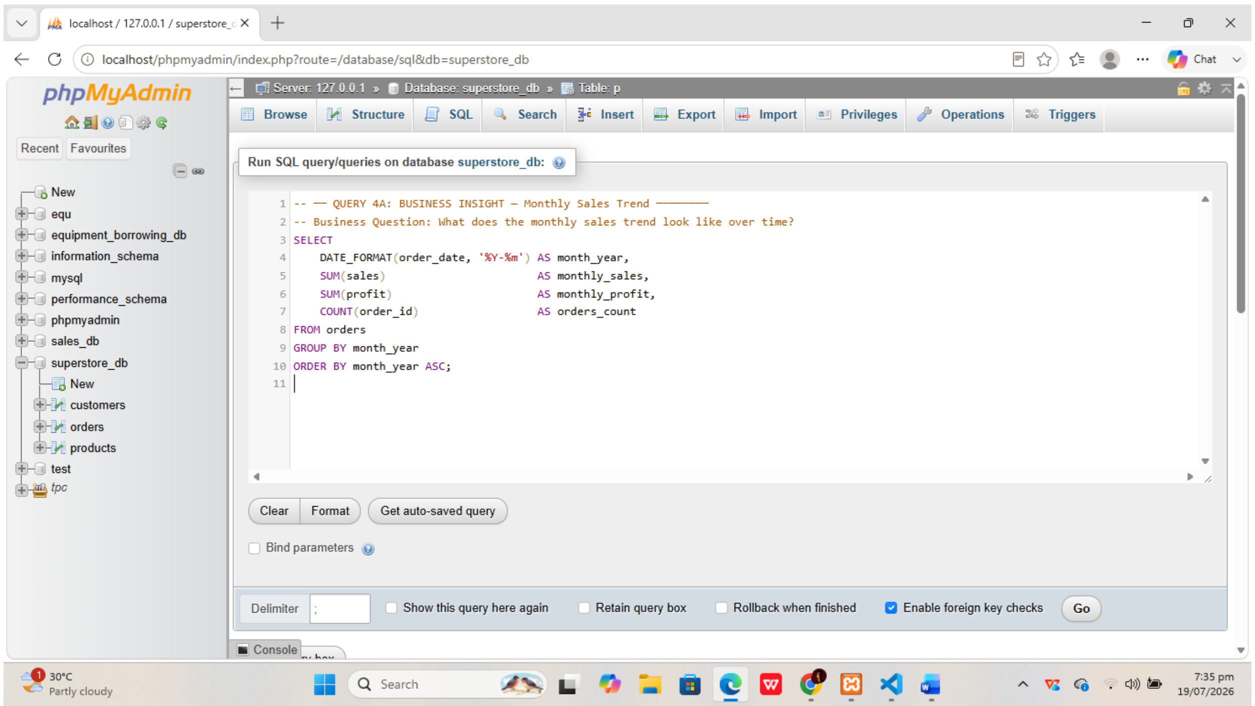
4. Business-Relevant Insights (2 Queries)

Queries specifically designed to answer your key business questions listed in Task 1.1.

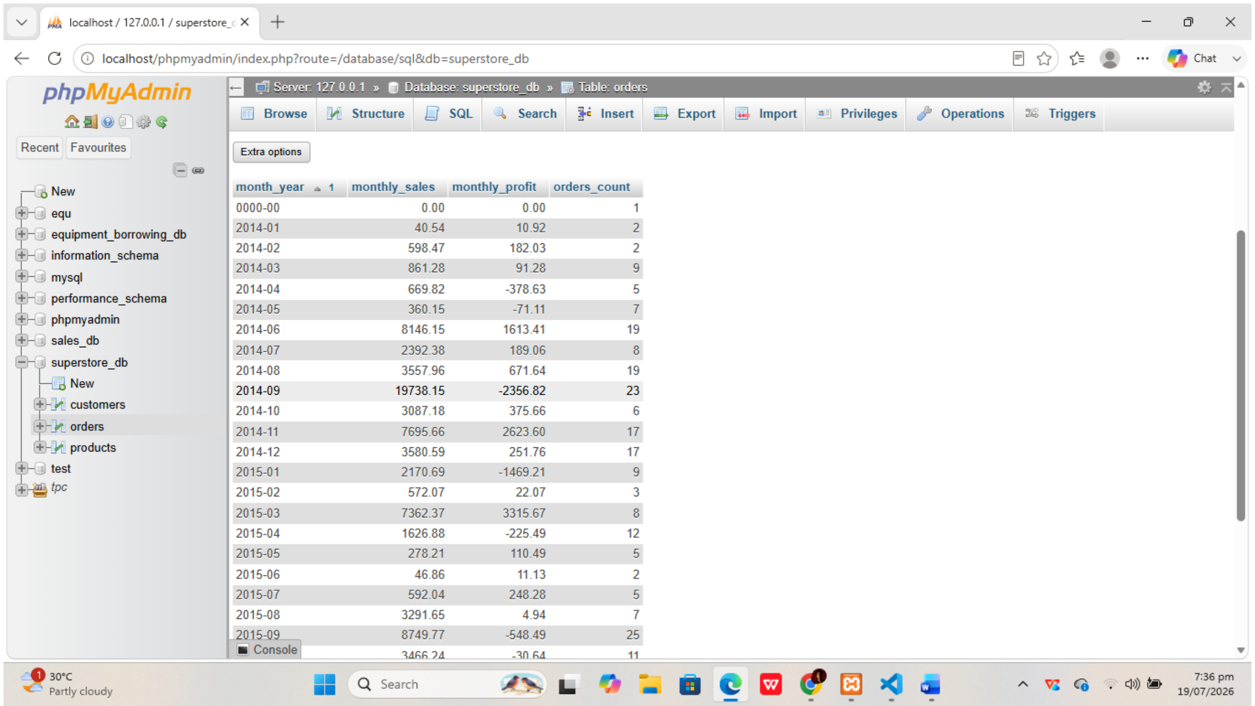
QUERY 4A: BUSINESS INSIGHT — Monthly Sales Trend (Question 3)

-- Business Question: What does the monthly sales trend look like over time?

- ***Screenshot of SQL Code***



- ***Result Screenshot***



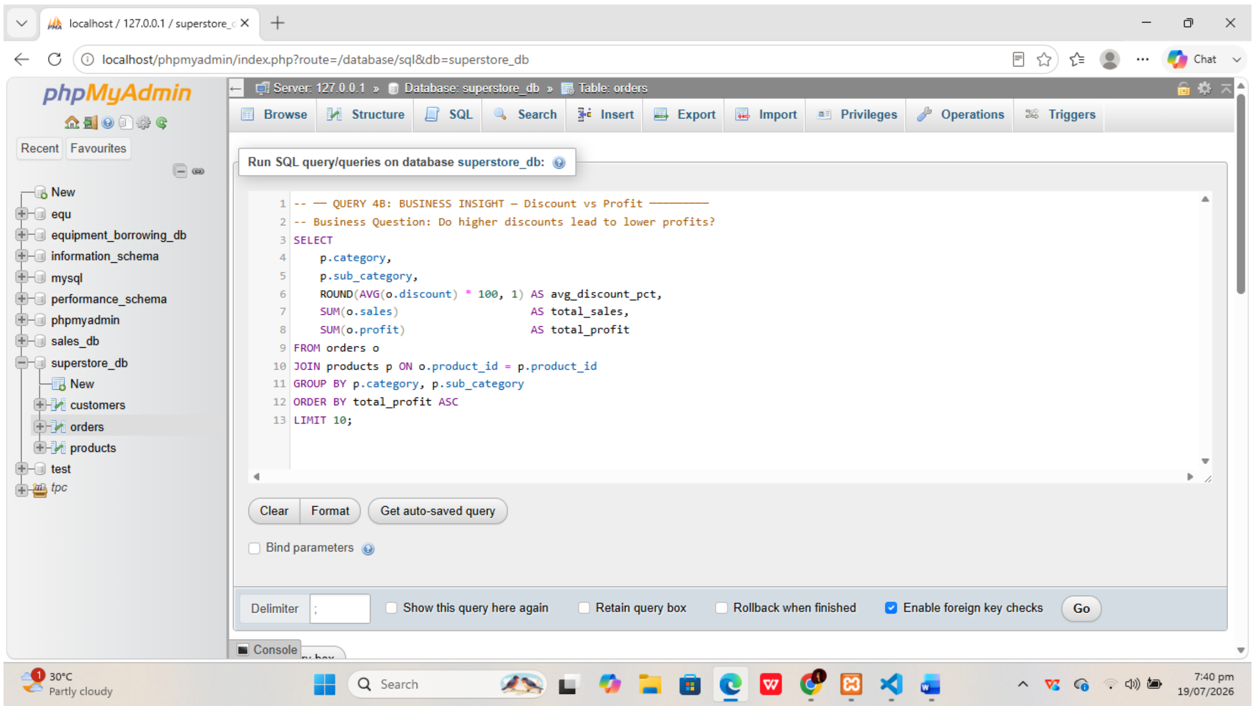
month_year	monthly_sales	monthly_profit	orders_count
0000-00	0.00	0.00	1
2014-01	40.54	10.92	2
2014-02	598.47	182.03	2
2014-03	861.28	91.28	9
2014-04	669.82	-378.63	5
2014-05	360.15	-71.11	7
2014-06	8146.15	1613.41	19
2014-07	2392.38	189.06	8
2014-08	3557.96	671.64	19
2014-09	19738.15	-2356.82	23
2014-10	3087.18	375.66	6
2014-11	7695.66	2623.60	17
2014-12	3580.59	251.76	17
2015-01	2170.69	-1469.21	9
2015-02	572.07	22.07	3
2015-03	7362.37	3315.67	8
2015-04	1626.88	-225.49	12
2015-05	278.21	110.49	5
2015-06	46.86	11.13	2
2015-07	592.04	248.28	5
2015-08	3291.65	4.94	7
2015-09	8749.77	-548.49	25
	3466.24	-30.64	11

- ***Business Interpretation***

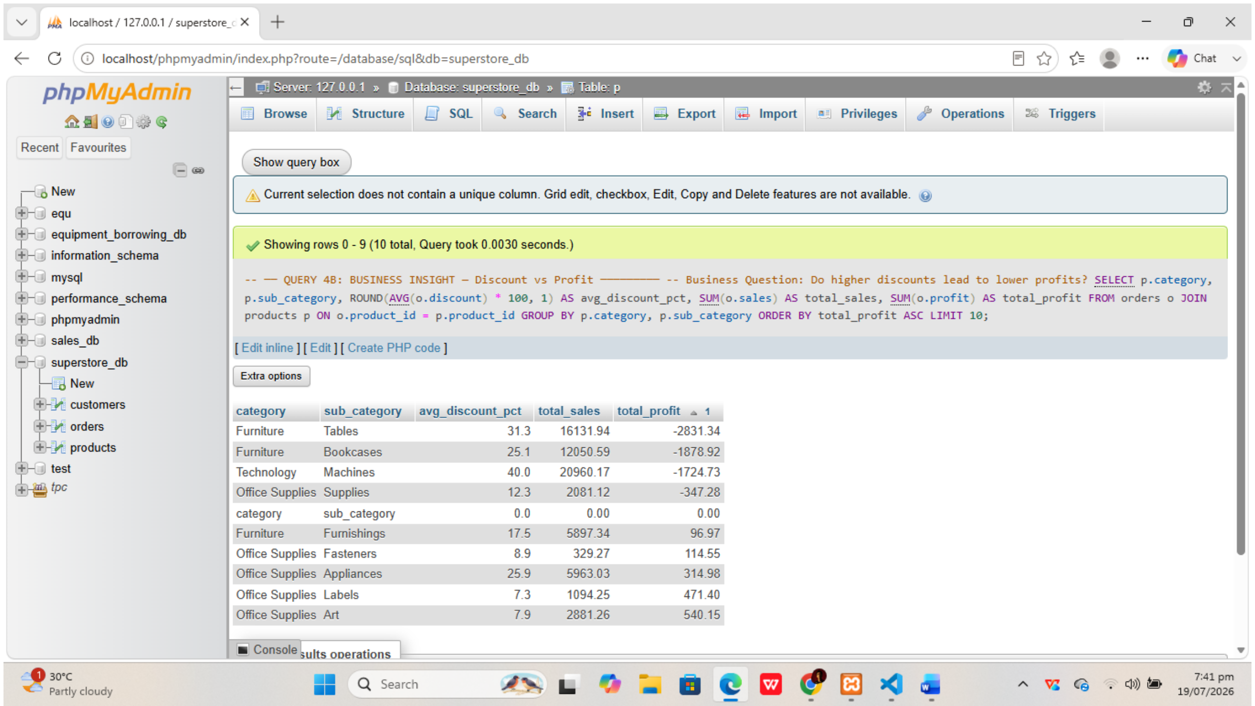
Query 4A: Groups total sales and profit by month to reveal peak sales periods such as the November–December holiday season and slower months. This time-series output is essential for demand forecasting and inventory planning for upcoming quarters, directly answering the business question about seasonal trends.

QUERY 4B: BUSINESS INSIGHT — Discount vs Profit (Question 2)
-- Business Question: Do higher discounts lead to lower profits?

- ***Screenshot of SQL Code***



- ***Result Screenshot***



- ***Business Interpretation***

Query 4B: Identifies the 10 least profitable sub-categories along with their average discount rates. Sub-categories with high average discounts and low or negative profit confirm that aggressive discounting is hurting margins, providing actionable data to revise the discounting strategy.

Time Finished – 19/07/2026, 7:43 pm